

vision**2**learn

Coding Practices Level 3

Unit 4

Learning
outcomes



Unit 4 - Software testing

Learning outcomes / Assessment criteria

1 Understand different levels, types and methods of testing

- 1.1 Identify different levels, types and methods of testing
- 1.2 Explain the types and methods of testing used within the stages of the Software Development Lifecycle
- 1.3 Explain the differences between functional and non-functional testing

2 Know about test driven development

- 2.1 Explain what is meant by test driven development
- 2.2 Describe the structural composition of a unit test
- 2.3 Explain the advantages and disadvantages of unit testing
- 2.4 Consider what would make a bad unit test

3 Be able to identify and fix a bug

- 3.1 Explain techniques which can be used to identify and fix bugs
- 3.2 Describe the benefits of the following techniques:
 - variable watching
 - compiler warnings
 - unit test
- 3.3 Diagnose the cause of a bug using the following techniques:
 - variable watching
 - compiler warnings
 - unit test
- 3.4 Perform a fix on broken code
- 3.5 Reflect on how the bug could have been prevented